

Post-Quantum Cryptography (PQC) Transition & Workforce Readiness

by Elijah Kim

1. Why the FAA Has to Move Now

Mandated Deadlines: The mandate is fixed. Executive Order 14412 sets 31 December 2030 for key establishment and 31 December 2031 for digital signatures. OMB M-26-15 turns that into five phases through 2035, with the agency migration plan, inventory methodology, and resource estimates due in October 2026.

Accruing Exposure: The exposure is already accruing. Traffic and records captured today can be decrypted once a cryptanalytically relevant quantum computer exists. Aviation assets serve twenty to thirty years, so equipment specified this year will still be flying when RSA and elliptic curve cryptography can no longer be trusted.

NAS Operational Integrity: For the National Airspace System (NAS), the exposure is integrity, not only confidentiality. Signature-dependent functions fail toward forgery and spoofing: loadable software part signing, CPDLC and ATN/IPS message integrity, SWIM data exchange, and any future ADS-B authentication. In an air traffic system, that is a critical safety question.

Agency Position & Roadmap: The FAA has already taken this position. RFI 697DCK-26-RFI-PQC calls PQC a foundational enabler of NAS modernization rather than a compliance exercise, sets a 2027 pilot ahead of rollout waves from 2028 to 2036, and names the technical workforce as one of three critical resources.

Compounding Delays: Delay compounds. Certification evidence under DO-178C and DO-254, AMS procurement cycles, and hardware refresh windows all run long. Crypto-agility has to be in the requirements before the first wave, because retrofitting it means paying the recertification bill twice.

KEY STRATEGIC TAKEAWAY

Crypto-agility must be built into specifications prior to the 2027 pilot wave. Retrofitting post-quantum cryptography later will double DO-178C / DO-254 recertification costs.

2. Why Training is the Binding Constraint

The algorithms are settled and vendors are shipping. What is scarce is people who can inventory what the agency actually runs, judge competing vendor claims, write defensible PQC requirements into a specification, carry a cryptographic change through a safety case, and operate the result for decades.

Untrained adoption produces predictable failures:

- **Compliance Risks:** Modules procured without FIPS 140-3 validation.

- **Configuration Drift:** Hybrid modes misconfigured so they quietly lose the protection they were bought for.
- **Architectural Lock-In:** Architectures created with no agility, leading to costly redesigns.
- **Vendor Dependencies:** Vendor lock-in that only becomes visible at the third wave.

Each failure mode is significantly cheaper to prevent through proactive technical training than to remediate through re-procurement and emergency system overhauls.

3. What Training is Actually Needed

Targeted curriculum mapping tailored across key FAA organizational roles:

Target Audience	Focus & Curriculum Scope	Depth & Duration
Executive & Acquisition Leadership	Mandate commitments, budget & resourcing consequences, integrated suite vs. modular build, and vendor evaluation frameworks.	Half day
NAS Operational Engineering	Latency budgets under Trajectory Based Operations (TBO), larger keys vs. MTU limits/fragmentation on legacy links, ADS-B, CPDLC, ATN/IPS, WAAS, and isolation for legacy non-migratable systems.	Two days
Enterprise IT & Security Operations	Cryptographic inventory and CBOM across COTS, custom, and SaaS estates; harvest-now triage by confidentiality lifespan; hybrid TLS 1.3 and IKEv2; PKI lifecycle management.	1 to 2 days
Safety & Certification Specialists	Cryptographic impact on DO-178C & DO-254 evidence at DAL A and B; agility as a core design requirement; alignment with RTCA SC-214, EUROCAE WG-67, and ICAO profiles.	One day
Sustainment & Field Technicians	Key management and rollover operations; monitoring and failure modes on PQC-enabled systems; incident response protocols for cryptographic components.	Recurring

4. How It Should Be Delivered

- **Capstone Table-Top Workshops:** Instructor-led workshops converging on a capstone table-top exercise that scores a realistic PQC proposal directly against the FAA's upcoming 2027 pilot.
- **Hands-On Inventory Clinics:** A hands-on inventory clinic run against a representative FAA system profile, ensuring participants leave with an actionable execution method rather than just slide decks.
- **Train-the-Trainer Scaling:** Establishing a internal train-the-trainer cadre to scale material across facilities through the 2028–2036 rollout waves, with refreshed modules tied to evolving standards and vendor maturity.

5. What Good Training Should Leave Behind

A successfully executed PQC workforce readiness initiative delivers the following key artifacts:

- **Scored Vendor & Architecture Rubric:** A standardized scoring tool to rigorously assess third-party PQC offerings and architectural proposals.
- **Cryptographic Inventory Methodology:** Standardized discovery and cataloging procedures covering both NAS operational technology (OT) and Enterprise IT.
- **Workforce-Readiness Annex:** A fully drafted readiness strategy ready for seamless insertion into the upcoming OMB M-26-15 migration submission.
- **Certified Internal Cadre:** A qualified, internal subject-matter team capable of guiding the decade-long transition without total vendor dependency.